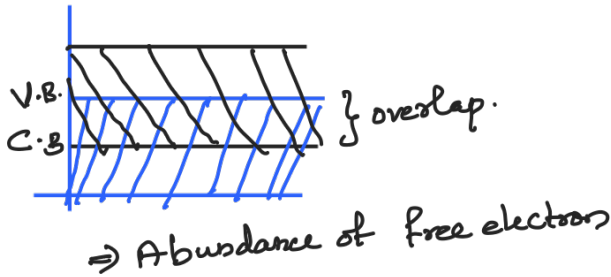
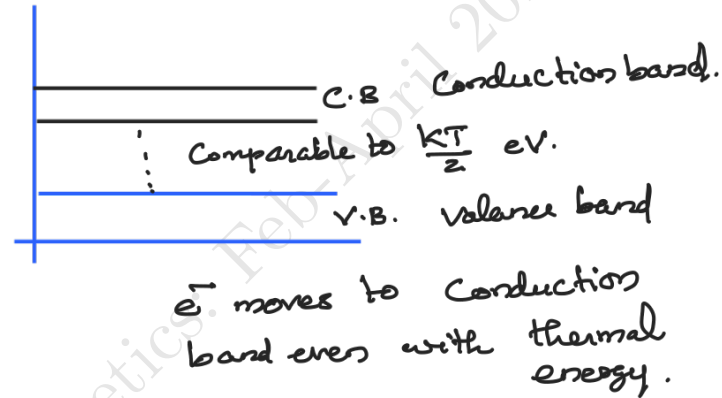
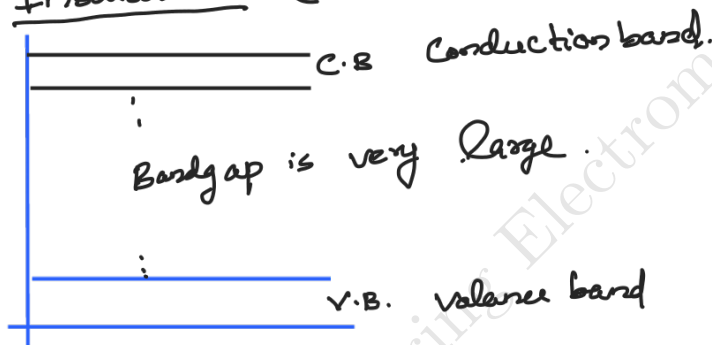


1. Electric field in dielectric medium.
2. Boundary conditions (dielectric-dielectric)

EE1204: Engineering Electromagnetics

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Office: EE514, aneesh@ee.iith.ac.inReview of Solid state physics.Conductor:Semiconductor:Insulators: (Dielectric)

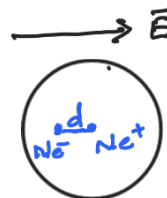
≈ Approximately equivalent to not having conduction band and e^- are bound in the valence band all in its place.

⇒ No free electrons.

When $\vec{E}$ field applied to dielectric atoms.

⇒ Its all orbitals gets distorted but e^- are not free to move out of the band. Thus only a small displacement in its position happens to the e^- cloud.

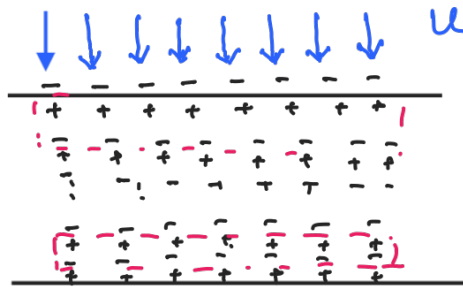
ie;



A dipole is induced!

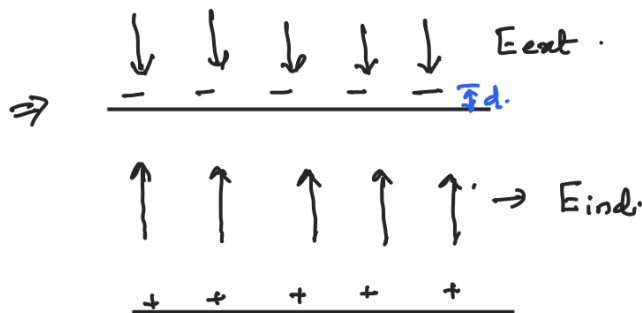
$N \rightarrow$ number of e^- in the atom.

Consider a dielectric block



Uniform electric field. $\vec{E}$

Internal dipole cancels out keeping -ve surface charge at the top and +ve at the bottom.



$$\vec{E}_{\text{ext}} = \vec{E}_{\text{external}}$$

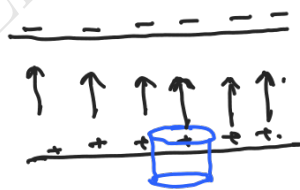
$$\vec{E}_{\text{ind}} = \vec{E}_{\text{induced}}$$

$$\sigma_{\text{ind}}^{-ve} \Rightarrow (-e)nd \rightarrow \text{number/m}^2. \text{ (surface charge density)}$$

$\hookrightarrow$ number density (number/m³)

$$\sigma_{\text{ind}}^{+ve} \Rightarrow (+e)nd.$$

Apply Gauss's law: (assuming only induced $\vec{E}_{\text{ind}}$ is present)



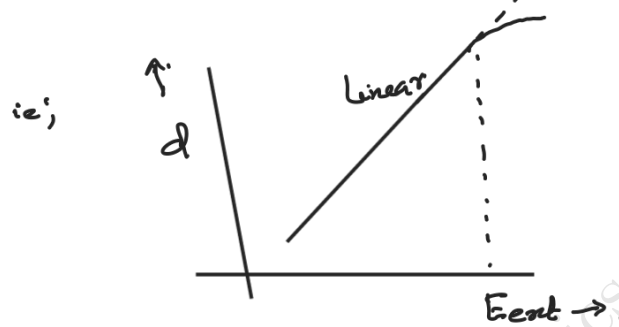
$$\vec{E}_{\text{ind}} \cdot \vec{A} = \frac{\sigma_{\text{ind}} \cdot A}{\epsilon_0} \rightarrow \text{Charge enclosed.}$$

$$\Rightarrow \vec{E}_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\epsilon_0}$$

$\vec{E}_{\text{ind}} \rightarrow$ is -ve as it is opposite to external applied field.

$$\text{ie; } E_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\epsilon_0} = \frac{(-e)nd}{\epsilon_0} \leftarrow \text{unknown?}$$

Experiments shows that as stronger the E_{ext} more is the e^- cloud shift..

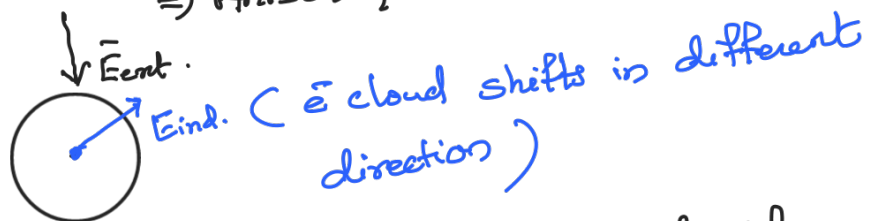


In most of the engineering application 'd' is linear with E_{ext}

$$\text{ie; } E_{\text{ind}} \propto E_{\text{ext}}.$$

Precautionary Note: There are materials where induced electric field may not be in the direction of E_{ext} .

=> Anisotropic material



But here we are considering isotropic material.

$$\vec{E}_{ind} = -\frac{e_{ind}}{\epsilon_0} = -\chi_e \vec{E}_{ext} =$$

Free charge under our control. induced charge density

$\chi_e \rightarrow$ electric susceptibility

$$\nabla \cdot \vec{E}_{ext} = \frac{\rho_{total}}{\epsilon_0} = \frac{\rho_f + \rho_b}{\epsilon_0} \quad \dots (1)$$

* When there is an abrupt change in $\vec{E}_{ind}$ (i.e. from 0 to $\vec{E}_{ind}$ or $\vec{E}_{ind}$ to 0) $\Rightarrow$ charge is induced.

* When $\vec{E}_{ind}$ is uniform no charge induced.

$$\text{i.e.; } \nabla \cdot \vec{E}_{ind} = \frac{\rho_b}{\epsilon_0} \quad \dots (2)$$

$$(1) - (2) \Rightarrow \nabla \cdot (\vec{E}_{ext} - \vec{E}_{ind}) = \frac{\rho_f}{\epsilon_0}$$

$$\nabla \cdot \epsilon_0 (\vec{E}_{ext} + \chi_e \vec{E}_{ext}) = \rho_f$$

$$\nabla \cdot [\epsilon_0 (1 + \chi_e) \vec{E}_{ext}] = \rho_f$$

Displacement vector $\vec{D}$

$\hookrightarrow$ Directly related to $\vec{E}_{ext}$
 $\hookrightarrow$ Related to material property.

Gauss's law in $\vec{D}$

$$\boxed{\nabla \cdot \vec{D} = \rho_f}$$

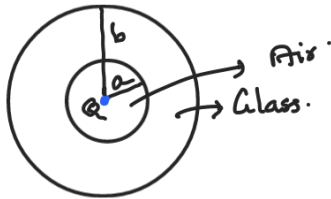
Gives $\vec{D}$ how to find $\vec{E}$?

The manufacture of the material provides

(experimentally) $1 + \chi_e \equiv \epsilon_r$ Relative permittivity or dielectric constant of the material. (Linear dielectric under gives $\vec{E}$)

$$\Rightarrow \vec{D} = \epsilon_0 \epsilon_r \vec{E} \Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon_0 \epsilon_r}.$$

Example: The usefulness of displacement vector concept.



Find the electric field and potential everywhere when the charge Q is placed at the center of spherical region (radius, a) filled with air!

For $r < a$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \cdot (\text{Coulomb's law})$$

For $a < r < b \rightarrow$ In glass $\vec{E}_{ind}$ is not known.

But, $\nabla \cdot \vec{D} = \rho_f$ --- Applicable to all ' r '.

Due to spherical symmetry. only $D_r = \vec{D} \cdot \hat{r}$ Component exist.

$$\Rightarrow \oint \vec{D} \cdot d\vec{s} = \iiint_V \nabla \cdot \vec{D} dv = \iiint_V \rho_f dv = Q$$

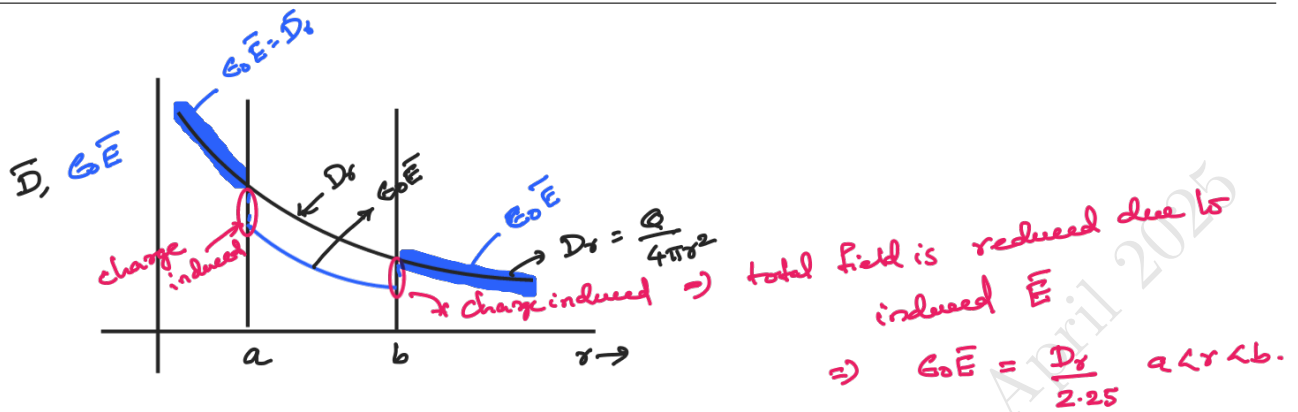
$$\Rightarrow 4\pi r^2 D_r = Q$$

$$\boxed{D_r = \frac{Q}{4\pi r^2}} \quad \checkmark$$

Now $\epsilon_0 \epsilon_{\text{glass}} E_r = D_r$

$$\epsilon_{\text{glass}} \approx 2.25.$$

$$E_r = \frac{D_r}{\epsilon_0 \epsilon_{\text{glass}}} \quad \text{for } a < r < b.$$



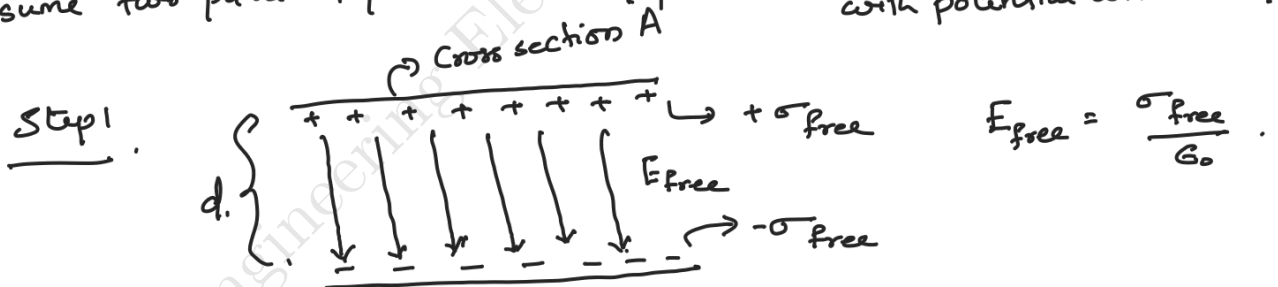
Now from $\vec{E}$ field we can calculate potential:

please verify by yourself.

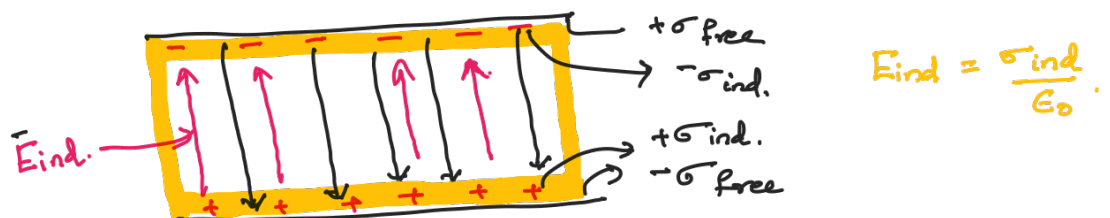
$$\phi = - \int_{\infty}^r E_r dr = \begin{cases} \frac{Q}{4\pi\epsilon_0 r} & ; r > b. \\ \frac{Q}{4\pi\epsilon_0 b} + \frac{Q}{4\pi\epsilon_0 \epsilon_{\text{glass}}} \left(\frac{1}{r} - \frac{1}{b} \right) & ; a < r < b \end{cases}$$

Revisiting Capacitor problem

Assume two parallel plates which are charged as below; with potential difference ϕ_0 .



Step 2. Now introduce a dielectric in between the plates:



$$\vec{E}_{\text{Net}} = \vec{E}_{\text{free}} + \vec{E}_{\text{ind}} \Rightarrow E_{\text{Net}} = E_{\text{free}} - E_{\text{ind}}$$

We know

$$\sigma_{\text{ind}} = \alpha \sigma_{\text{free}} ; E_{\text{ind}} = \alpha E_{\text{free}}$$

$$\Rightarrow E_{\text{Net}} = E_{\text{free}} (1 - \alpha)$$

$$= \frac{E_{\text{free}}}{\epsilon_r} \longrightarrow (\text{dielectric constant})$$

ie, $\boxed{E_{\text{Net}} = \frac{E_{\text{free}}}{\epsilon_r}}$ Net field reduced by ϵ_r .

$\Rightarrow$ Now as $E_{\text{Net}} \downarrow \Rightarrow \Phi = E d$ (Denoting E_{Net} as E)
 $\Rightarrow \Phi \downarrow$ when we introduce dielectric.

* Natural dipoles have very large dielectric constant compared to that of induced dipole molecules.

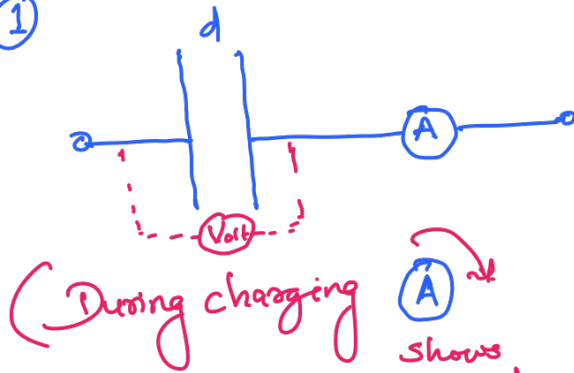


e^- spends more time towards oxygen than that to Hydrogen ($\Rightarrow$ Natural dipole,)
 and it has $\epsilon_r \approx 80$ (at room temperature)

Understanding Capacitance further:



①



* Charging this capacitor by connecting to power supply is 10V.

$$d = 1\text{mm}; \phi = 10\text{V}$$

Step 1

* After charging disconnect the power supply

Step 2: Increase $d = 10\text{mm}$.

$Q_{\text{free}} \rightarrow$ No change.

$\Rightarrow$ Amp-meter does not show anything.

$\vec{E} \rightarrow$ No change.

potential: $\phi = E d \Rightarrow \phi \uparrow 10 \text{ times.} \Rightarrow \phi = 100\text{V}$

②

$\phi = 100\text{V}; d = 10\text{mm};$ Insert dielectric with ' ϵ_r '.

$Q_{\text{free}} \rightarrow$ No change. $\Rightarrow$ (A) $\rightarrow$ meter does nothing.

But due to induced $\vec{E}$ $\rightarrow E \downarrow \epsilon_r \text{ times}$ $\Rightarrow \phi \downarrow \epsilon_r \text{ times.}$ $\uparrow C_{\text{new}} = \frac{Q_{\text{free}}}{\phi \downarrow} \Rightarrow C_{\text{new}} = \frac{\epsilon_0 \epsilon_r A}{d}$



$$E = \frac{\sigma_{\text{free}}}{\epsilon_0 \epsilon_r} \dots (1)$$

$$\phi = Ed \dots (2)$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{Q_{\text{free}}}{\phi} \dots (3)$$

For capacitors.

③ Charge Capacitor ^(air filled) with 10 V (But this time not disconnecting the power supply)

$$\phi_0 = 10 \text{ V}; \quad d = 1 \text{ mm} \quad \textcircled{A} \downarrow$$

Now increase d to 10 mm.

* Now, potential difference is fixed irrespective of d as power supply is still connected. $\Rightarrow \phi = \phi_0 = 10 \text{ V}$

• As d increases; From Eqn (2) $E \downarrow 10$ times to keep $\phi = \phi_0$.

* $\epsilon \downarrow 10$ times ($\downarrow C = \frac{A \epsilon_0}{d \uparrow}$)

$$\text{Now } \downarrow C = \frac{Q_{\text{free}}}{\phi} = \frac{(Q_{\text{free}}) \downarrow 10 \text{ times}}{\phi_0} \Rightarrow Q_{\text{free}} \text{ should reduce by } 10 \text{ times}$$

$\Rightarrow$ Charge will flow away from the plates $\Rightarrow \textcircled{A} \uparrow$ shows reverse direction.

④ Now introduce dielectric: But power supply still connected.

$\Phi = 10 \text{ V}$, $d = 10 \text{ mm}$: Introduce ' ϵ_r ' (dielectric)

$\Rightarrow \Phi = \Phi_0$ unchanged. (as connected with power supply)

(a) Q_{free} increases
by factor of ϵ_r .

$$\therefore C = \frac{\epsilon_r \epsilon_0 A}{d} \Rightarrow C \uparrow \epsilon_r \text{ times}$$

$$\Rightarrow \frac{Q_{\text{free}} (\uparrow \epsilon_r \text{ times})}{\Phi_0}$$



$\Rightarrow$ Charge goes into plates.

Here $\Phi_0 = \text{Constant}$
 $d = \text{Constant}$

$\Rightarrow$ ' E ' does not change (even after introducing dielectric)

$$E = \Phi_0 d.$$

* ' E ' does not change even after introducing ' ϵ_r ' dielectric.

This is why Q_{free} has to increase.

Because dielectric introduces induced Electric field and hence to compensate reduction in E field nature has to incorporate more free charges & hence $Q_{\text{free}} \uparrow$